

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts soil and weather parameters from users	Frontend	Done	High
2	Crop Prediction	Validates user input before prediction	Backend	Done	High
3	Result Display	Predicts the most suitable crop using the ML model	ML Model	Done	High
4	Responsive User Interface	Displays the predicted crop with a description	Frontend	Done	High
5	Model Integration	Provides a clean and user-friendly interface	UI	Done	High
6	Model Integration	Integrates the trained ML model with Flask	Backend	Done	High
7	Navigation	Enables smooth navigation between pages	Frontend	Done	High
8	Error Handling	Handles invalid inputs and prediction errors	Backend	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page
2	Backend / Logic	3	Crop Prediction, Flask Routing, Error Handling
3	Database / Storage	2	Crop Dataset (CSV), Trained Model (.pkl)
4	API / Integration	1	Flask-ML Model Integration
5	Security / Authentication	1	Input Validation